Chapter 8

Direct Methods for Stability Analysis: An Introduction

8.1 INTRODUCTION

The conventional time–domain approach numerically integrates both the fault-on system and the postfault system. The stability of the postfault system is assessed based on the simulated postfault trajectory. The typical simulation period for the postfault system is 10 s and can go beyond 15 s if multiswing instability is of concern, making this conventional approach rather time-consuming. This time-consuming nature makes the conventional time–domain approach infeasible for practical application in power system online stability assessments (Chiang, 1996, 1999; El-kady et al., 1986; Groom et al., 1996). By contrast, direct methods only integrate the fault-on system and determine, without integrating the postfault system, whether or not the postfault system will remain stable once the fault is cleared by comparing the system energy (when the fault is cleared) to a critical energy value. Direct methods not only avoid the time-consuming numerical integration of the postfault system but they also provide a quantitative measure of the degree of system stability/instability (Chiang, 1991; Chiang et al., 1995; Fouad and Vittal, 1988; Gibescu et al., 2005; Hiskens and Hill, 1989).

The fundamental problem of transient stability is the following: starting from the postfault initial state $X(t_{cl})$, will the postfault system settle down to the steady state condition X_s? In other words, the purpose of power system stability analysis is to determine whether the initial point of the postfault trajectory is located inside the stability region (domain of attraction) of an acceptable stable equilibrium point (SEP) (acceptable steady state).

The problem of direct stability analysis can be translated into the following: given a set of nonlinear equations with an initial condition, determine whether or not the ensuing trajectories will settle down to a desired steady state without resorting to explicit numerical integrations. It is assumed in the area of direct methods that the following condition is satisfied:

Direct Methods for Stability Analysis of Electric Power Systems, by Hsiao-Dong Chiang

Assumption (A1): The prefault SEP, X_s^{pre}, lies inside the stability region of a desired postfault SEP, X_s.

At present, the most popular method for analyzing transient stability is to numerically simulate the postfault dynamic behaviors via numerical integrations of the postfault system model. On the other hand, direct methods first assume that the postfault SEP, X_s, which satisfies (power system) operational constraints (an acceptable steady state), exists. If this point does not exist, then the direct method cannot solve the problem and must resort to the time–domain simulation for verification. Direct methods next determine whether the initial point of the postfault trajectory lies inside the stability region of the acceptable SEP. If it does, direct methods then declare that the resulting postfault trajectory will converge to X_s without knowing the dynamics of the postfault trajectory.

The basis of direct methods for the stability assessment of a postfault system is knowledge of the stability region: if the initial condition of the postfault system lies inside the stability region of a desired postfault SEP, then one can ensure, without performing any numerical integration, that the ensuring postfault trajectory will converge to the desired SEP. Therefore, knowledge of the stability region plays an important role in direct methods.

In the next three sections, a heuristic introduction to direct methods, including the closest UEP method, the potential energy boundary surface (PEBS) method, and the controlling UEP method will be presented.

8.2 A SIMPLE SYSTEM

Heuristic arguments for the applicability of direct methods can be derived from the classical equal-area criterion. We consider one-machine infinite bus systems described by the following equations:

$$\begin{aligned}\dot{\delta} &= \omega \\ M\dot{\omega} &= -D\omega - P_0 \sin\delta + P_m.\end{aligned} \tag{8.1}$$

There are three equilibrium points lying within the range of $\{(\delta, \omega) = -\pi < \delta < \pi, \omega = 0\}$, and they are $(\delta_s, 0) = (\sin^{-1}(P_m/P_0), 0)$, which is a SEP, and $(\delta_1, 0) = (\pi - \sin^{-1}(P_m/P_0), 0)$ and $(\delta_2, 0) = (-\pi - \sin^{-1}(P_m/P_0), 0)$, which are unstable equilibrium points (UEPs). We consider the following function, an energy function for this simple system:

$$E(\delta, \omega) = \frac{1}{2}M\omega^2 - P_m\delta - P_0\cos\delta. \tag{8.2}$$

The energy function can be divided into kinetic energy $K(\omega)$ and potential energy functions $U(\omega)$:

$$E(\delta, \omega) = K(\omega) + U(\delta), \tag{8.3}$$

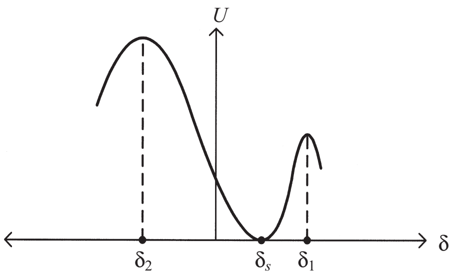

Figure 8.1 The potential energy function $U(\delta)$ is a function of δ only and reaches its local maximum at UEPs δ_1 and δ_2.

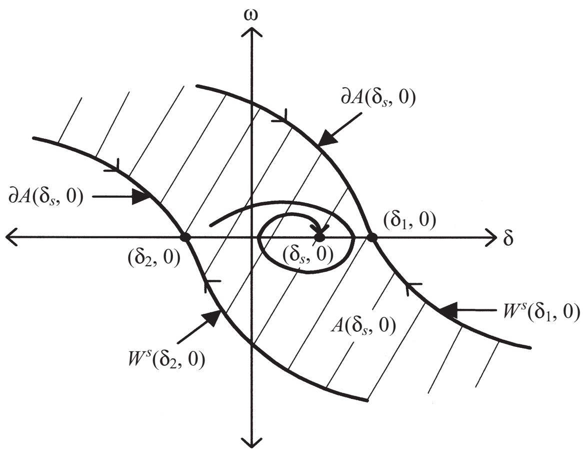

Figure 8.2 The position of the stable equilibrium point $(\delta_s, 0)$ along with its stability $A(\delta_s, 0)$ (the shaded area). The stability boundary $\partial A(\delta_s, 0)$ is composed of the stable manifold of the UEP $(\delta_1, 0)$ and the stable manifold of the UEP $(\delta_2, 0)$.

where $K(\omega) = \frac{1}{2}M\omega^2$ and $U(\delta) = -P_m\delta - P_0\cos\delta$. The potential energy function $U(.)$ as a function of δ is shown in Figure 8.1. We notice that function $U(\delta)$ reaches its local maximum at UEPs δ_1 and δ_2 of the following (reduced) system:

$$\dot{\delta} = -P_0 \sin\delta + P_m. \tag{8.4}$$

The system is two-dimensional (2-D). Hence, the stability region of $(\delta_s, 0)$, shown in Figure 8.2, is 2-D. Regarding the stability region $A(\delta_s, 0)$, we have the following:

1. The exact stability region $A(\delta_s, 0)$ is completely characterized by its stability boundary $\partial A(\delta_s, 0)$, which is composed of the stable manifold of the UEP $(\delta_1, 0)$ and the stable manifold of the UEP $(\delta_2, 0)$ (see Figure 8.2).
2. The intersection between the stability region $A(\delta_s, 0)$ and the angle space $\{(\delta, \omega): \delta = R, \omega = 0\}$ is $A_\delta := \{(\delta, \omega): \delta \in [\delta_2, \delta_1], \omega = 0\}$.
3. The boundary of this one-dimensional region, A_δ, is composed of two points, δ_1 and δ_2, where $(\delta_1, 0)$ and $(\delta_2, 0)$ are the UEPs on the stability boundary $\partial A(\delta_s, 0)$.

4. These two points, δ_1 and δ_2, are characterized as being the local maxima of the potential energy function $U(.)$.

The stability for this simple system can be directly assessed on the basis of the energy function $U(\delta)$, without the knowledge of the postfault trajectory. The methods presented in the next three sections can be employed to assess the system's stability:

8.3 CLOSEST UEP METHOD

The closest UEP method is the classical direct method developed in the late 1960s. This method uses the constant energy surface $\{(\delta, \omega): V(\delta, \omega) = U(\delta_1)\}$, passing through the closest UEP $(\delta_1, 0)$ to approximate the stability boundary $\partial A(\delta_s, 0)$. Regarding the simple system discussed in the previous section, we have the following observation:

- Of the exact stability boundary $\partial A(\delta_s, 0)$, the UEP $(\delta_1, 0)$ has the lowest energy function value among all the UEPs on the stability boundary $\partial A(\delta_s, 0)$. Hence, $(\delta_1, 0)$ is termed the closest UEP of $(\delta_s, 0)$ with respect to the energy function $U(\delta)$.

If a given state, say, $(\delta_{cl}, \omega_{cl})$, whose energy function value $V(\delta_{cl}, \omega_{cl})$ is less than $U(\delta_1)$, then the state $(\delta_{cl}, \omega_{cl})$ is classified to be lying inside the stability region of $(\delta_s, 0)$ (see Figure 8.3). Thus, one can assert without numerical integration that the resulting trajectory will converge to $(\delta_s, 0)$.

This closest UEP method, although simple, can give considerable conservative stability assessments, especially for those fault-on trajectories crossing the stability boundary $\partial A(\delta_s, 0)$ through $W^s(\delta_2, 0)$ (see Figure 8.4a). For example, the postfault trajectory starting from state P, which lies inside the stability region $A(\delta_s, 0)$, is classified to be unstable by the closest UEP method. This is incorrect because the resulting trajectory will converge to $(\delta_s, 0)$ and hence it is stable (see Figure 8.4b).

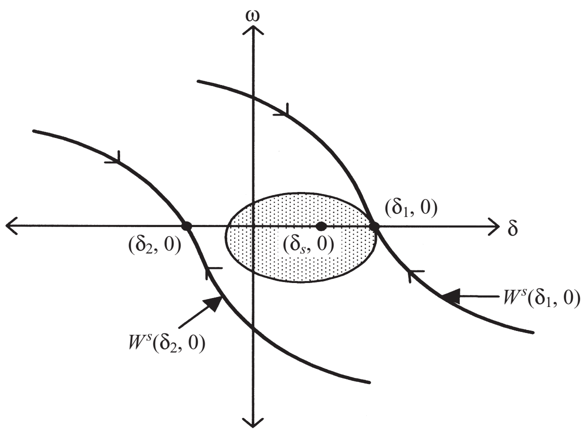

Figure 8.3 The closest UEP method uses the constant energy surface passing through the closest UEP $(\delta_1, 0)$ to approximate the (entire) stability boundary $\partial A(\delta_s, 0)$. The shaded area is the estimated stability region by the closest UEP method.

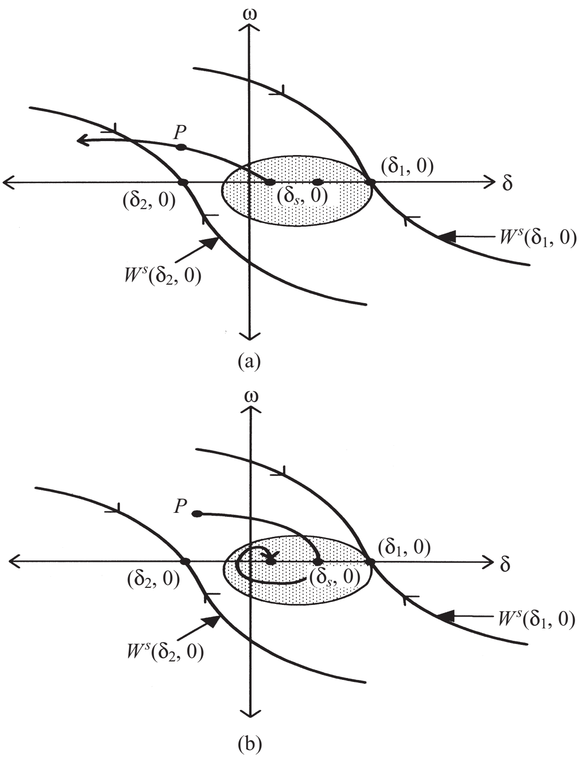

Figure 8.4 (a) The closest UEP method gives considerable conservative stability assessments for those fault-on trajectories crossing the stability boundary $\partial A(\delta_s, 0)$ through $W^s(\delta_2, 0)$. (b) The postfault trajectory starting from state P, which lies inside the stability region $A(\delta_s, 0)$, is classified to be unstable by the closest UEP method, while in fact the resulting trajectory will converge to $(\delta_s, 0)$ and hence it is stable.

The closest UEP method does provide an accurate approximation for the entire stability boundary. However, it does not provide an accurate approximation for the relevant stability boundary. This is due to the fact that the fault-on trajectory is not taken into account in the closest UEP method. The conservativeness of the closest UEP method is now well recognized. Because of this severe conservativeness, the closest UEP method is not popular as a direct method in practical applications. A rigorous analysis of the closest UEP method along with its theoretical foundation will be presented in Chapter 9.

8.4 CONTROLLING UEP METHOD

The controlling UEP method, developed in the 1980s, aims to reduce the conservativeness of the closest UEP method by taking the dependence of the fault-on trajectory into account. We illustrate this method using the simple system as an example. To assess the stability of the postfault trajectory whose corresponding fault-on trajectory $(\delta(t), \omega(t))$ moves towards δ_1, the controlling UEP method uses the constant energy surface passing through the UEP $(\delta_1, 0)$, which is $\{(\delta, \omega): E(\delta, \omega) = U(\delta_1)\}$ as the local approximation for the relevant stability boundary of the postfault system.

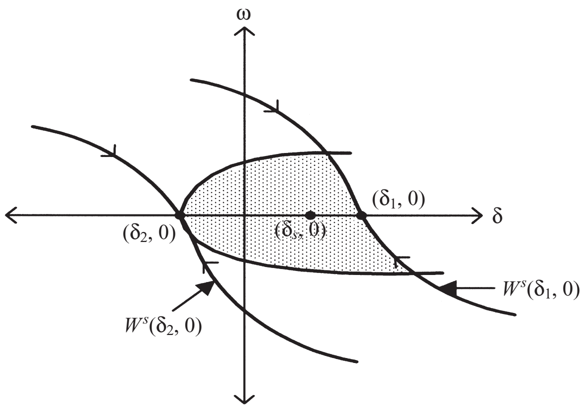

Figure 8.5 The controlling UEP method uses the constant energy surface passing through the controlling UEP to approximate the relevant stability boundary. The shaded area is the estimated stability region by the controlling UEP method. The controlling UEP method does not provide an approximation for the entire stability boundary. However, it provides an accurate approximation for the relevant stability boundary.

In the same manner, for those fault-on trajectories $(\delta(t), \omega(t))$ whose $\delta(t)$ component moves towards δ_2, the controlling UEP method uses the constant energy surface passing through the UEP $(\delta_2, 0)$, which is $\{(\delta, \omega): E(\delta, \omega) = U(\delta_2)\}$ as the local approximation for the relevant stability boundary (see Figure 8.5). Therefore, for each fault-on trajectory, there exists a unique corresponding UEP whose stable manifold constitutes the relevant stability boundary. This unique UEP is the controlling UEP.

The constant energy surface passing through the controlling UEP can be used to accurately approximate the relevant part of the stability boundary towards which the fault-on trajectory is heading. If the energy function value of a given state is less than that of the controlling UEP, then the state is classified as lying inside the stability region of $(\delta_s, 0)$ by the controlling UEP method. Thus, one can assert without numerical integration that the resulting trajectory will converge to $(\delta_s, 0)$.

The controlling UEP method uses the constant energy surface passing through the controlling UEP to approximate the relevant stability boundary. Apparently, the controlling UEP method does not provide an approximation for the entire stability boundary. However, it provides an accurate approximation for the relevant stability boundary. The controlling UEP method, although more complex than the closest UEP method, gives much more accurate and less conservative stability assessments than the closest UEP method. We illustrate this point on Figure 8.6. Suppose a fault is cleared when the fault-on trajectory reaches the state $(\bar{\delta}, \bar{\omega})$ in Figure 8.6. The postfault trajectory starting from the state $(\bar{\delta}, \bar{\omega})$, which lies inside the stability region $A(\delta_s, 0)$, will converge to the SEP $(\delta_s, 0)$. Hence, the postfault system is stable. This postfault trajectory is correctly classified as stable by the controlling UEP method, while it is classified to be unstable by the closest UEP method. Again, this

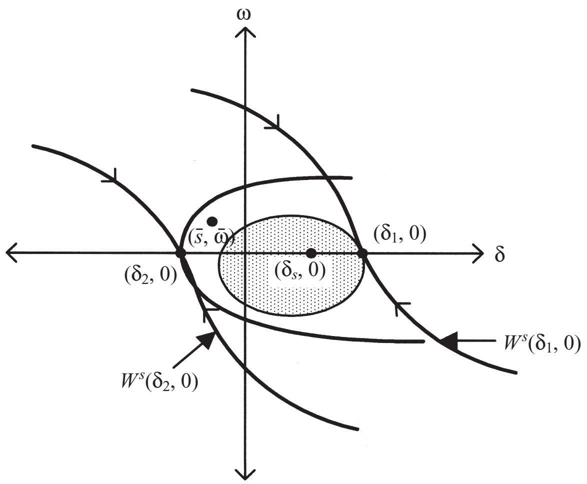

Figure 8.6 The postfault trajectory starting from the state $(\bar{\delta}, \bar{\omega})$, which lies inside the stability region $A(\delta_s, 0)$, is correctly classified as stable by the controlling UEP method, while it is classified to be unstable by the closest UEP method.

shows the conservativeness of the closest UEP method, which does not take into account the dependence of the fault-on trajectory.

The following three-step procedure is the basis of the controlling UEP method:

Step 1. Integrate the fault-on trajectory $(x(t),\dot{x}(t))$. Suppose the fault-on trajectory $(x(t),\dot{x}(t))$ intersects the stable manifold of a UEP lying on the stability boundary of the postfault SEP.

Step 2. Let the potential energy value of the UEP (i.e. the controlling UEP) be v. Then, the critical energy value for the fault-on trajectory $(x(t),\dot{x}(t))$ is v.

Step 3. Use the constant energy surface $\{(x,\dot{x}):V(x,\dot{x}) < v\}$ as a local approximation of the relevant stability boundary. If the energy value when the fault is cleared is less than the critical energy value v, then the postfault system will be stable; otherwise, it may be unstable.

A rigorous analysis of the controlling UEP method, along with its theoretical foundation, will be presented in Chapters 11 through 13.

8.5 PEBS METHOD

The main problem with the controlling UEP method is that it is generally very challenging to find the controlling UEP relative to a fault-on trajectory. The PEBS method attempts to find a local approximation of the relevant stability boundary without the need for computing the UEPs of the postfault system.

A heuristic explanation of the PEBS method using the one-machine infinite bus system is presented below:

1. From the potential energy function V_p (.), find the local maxima x_1 and x_2.

2. For those fault-on trajectories $(x(t),\dot{x}(t))$ whose $x(t)$ component moves towards x_1, the set, that is, the constant energy surface, $\{(x,\dot{x}):V(x,\dot{x}) = V_p(x_1)\}$ is chosen as the local approximation of the (relevant) stability boundary of $(\delta_s, 0)$. For those fault-on trajectories $(x(t),\dot{x}(t))$ whose $x(t)$ component moves towards x_2, the set $\{(x,\dot{x}):V(x,\dot{x}) = V_p(x_2)\}$ is chosen as the local approximation of the (relevant) stability boundary of $(\delta_s, 0)$.

It should be pointed out that for the one-machine infinite bus system, (1) the intersection between the stability boundary and the set $\dot{x} = 0$ is determined solely by the potential energy function $V_p(x)$, and (2) for this special case, the PEBS method behaves like the controlling UEP method.

Kakimoto et al. suggest that a "PEBS" be formed by joining these V_p maxima through the "ridge" of the potential energy level surface (Kakimoto et al., 1978). The PEBS constructed this way is clearly orthogonal to the equipotential curve $V_p(.)$. Furthermore, along the direction orthogonal to the PEBS, the potential energy $V_p(.)$ achieves a local maximum at the PEBS. The following two-step procedure was proposed as the basis of the PEBS method:

Step 1. Integrate the fault-on trajectory $(x(t),\dot{x}(t))$. Suppose $x(t)$ crosses PEBS at an intersection point whose potential energy value is v. The critical energy value for the fault-on trajectory $(x(t),\dot{x}(t))$ is v.

Step 2. Use the constant energy surface $\{(x,\dot{x}):V(x,\dot{x}) < v\}$ as a local approximation of the relevant stability boundary. If the energy value when the fault is cleared is less than the critical energy value v, then the postfault system will be stable; otherwise, it may be unstable.

The derivation of the PEBS method for the multimachine case is based on heuristic arguments. Theoretical justification for the PEBS procedure, except for the one-machine infinite bus case, is lacking. It is not clear, for the multimachine case, whether the PEBS is the intersection of the stability boundary of the multimachine system with the subspace $\{(x,\dot{x}):\dot{x} = 0\}$. Moreover, because of the heuristic way the PEBS is derived, it is difficult to check whether the method provides a good local approximation of the stability boundary or under what conditions the approximation is good. A rigorous analysis of the PEBS method, along with its theoretical foundation, will be presented in Chapter 10.

8.6 CONCLUDING REMARKS

The presented exposition on the one-machine infinite bus system outlines the key bases for the closest UEP method, the controlling UEP method, and the PEBS method. In addition, it has highlighted the differences between the direct stability assessments among these three methods. The above exposition also reveals the following three main steps needed for direct methods:

Step 1. Construct an energy function for the postfault system, say, $V(\delta, \omega)$.

Step 2. Compute the critical energy value V_{cr} for a given fault-on trajectory (say, based on the controlling UEP method).

Step 3. Compare the energy value of the state when the fault is cleared, say, $V(\delta_{cl}, \omega_{cl})$, with the critical energy value V_{cr}. If $V(\delta_{cl}, \omega_{cl}) < V_{cr}$, the postfault trajectory will be stable. Otherwise, it may be unstable.

Extending the above justification for direct methods to general multimachine power systems is nontrivial. This is partly due to the intrinsic differences between the nonlinear dynamics of 2-D systems and that of higher-dimensional nonlinear systems. Rigorous analysis and a theoretical foundation of direct methods will be presented in the following chapters. In particular, we will apply and extend the energy function theory to power system transient stability models to develop theoretical foundations for direct methods including the closest UEP method, the controlling UEP method, and the PEBS method. The focus of this book will be on the controlling UEP method since it is the best direct method.

In the context of direct methods for power system transient stability analysis, we point out one challenging task: given a point in the state space (say, the initial point of the postfault system), it is generally difficult to determine which connected component of a level set contains the point by simply comparing the energy at the given point and the energy of the level set. This is due to the fact that a level set usually contains several connected components and these components are not easy to differentiate based on an energy function value. For instance, the two states, x_1 and x_2, have the same energy function value. However, state x_1 lies outside the stability region, while state x_2 lies inside the stability region (see Figure 8.7 for the closest UEP method and Figure 8.8 for the controlling UEP method).

Fortunately, in the context of direct methods, this difficulty can be circumvented using direct methods since they compute the relevant pieces of information regarding (1) a prefault SEP, (2) a fault-on trajectory, and (3) a postfault SEP. These pieces of information are sufficient to identify the connected component of a level set that contains the initial point of the postfault system.

The above analysis also leads to the following fundamental assumption required for direct methods:

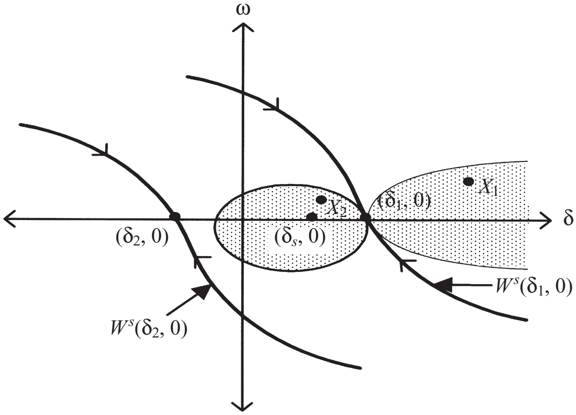

Figure 8.7 The closest UEP method uses the connected constant energy surface passing through the closest UEP $(\delta_1, 0)$ and containing the SEP to approximate the entire stability boundary $\partial A(\delta_s, 0)$. However, it is difficult to differentiate the two states highlighted solely on the basis of energy function values.

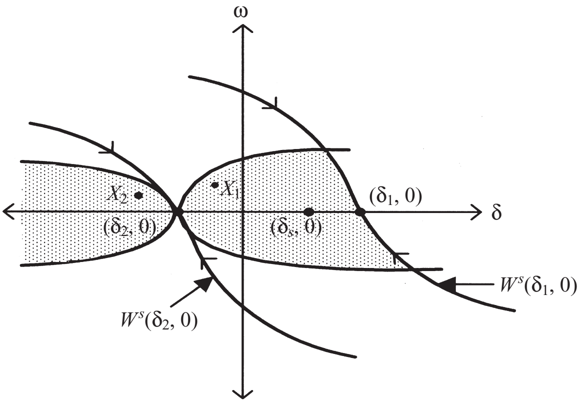

Figure 8.8 The controlling UEP method uses the constant energy surface passing through the controlling UEP and containing the SEP to approximate the relevant stability boundary. However, it is difficult to differentiate the highlighted two states solely on the basis of energy function values.

Fundamental Assumption: The prefault SEP lies inside the stability region of the postfault SEP.

This assumption is trivial if the prefault system equals the postfault system. Since the stability region is an open set of a certain size, this assumption is plausible for a great majority of contingencies. When this fundamental assumption is not satisfied for certain very extreme contingencies, then the time–domain simulation approach must be used for the stability analysis of these contingencies.